

Wei Zhou

Associate Professor in Perceptual Multimedia Computing, IEEE Senior Member

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Profile for IEEE Transactions on Multimedia Associate Editor Nomination

Research profile. Multimedia computing, human perception, and artificial intelligence, with particular strengths in image/video quality assessment, immersive and 3D media, AI-generated content quality, multimodal learning, and human-centric multimedia intelligence.

Eligibility and impact. PhD awarded in June 2021; more than five years post-PhD by September 2026. Google Scholar: 4,300+ citations; h-index 36. Based in the United Kingdom, with prior research and industry experience in Germany, Canada, Japan, China, Microsoft Research, Alibaba, and Intel.

Current Appointment and Research Leadership

- **2026–present** Associate Professor, Cardiff University, United Kingdom.
- **2023–2026** Assistant Professor, Cardiff University, United Kingdom.
- **Research leadership:** General Chair, IEEE QoMEX 2026 (first edition held in the UK); Steering Committee Member, QoMEX; member of the Visual Computing Research Section at Cardiff.
- **Professional service:** Committee Member, British Standards Institution; Member, EPSRC Peer Review College.

Editorial Experience

- **Associate Editor, IEEE Transactions on Neural Networks and Learning Systems** (2024–present).

IEEE / ACM Multimedia Community Leadership

- Elected Member, IEEE CASS Multimedia Systems and Applications Technical Committee.
- Elected Member, IEEE SPS Multimedia Signal Processing Technical Committee.
- Elected Member, EURASIP Visual Information Processing Technical Committee.
- **Conference leadership:** Senior Area Chair, IEEE ICME 2026; Area Chair, ACM Multimedia 2024–2026, IEEE ICIP 2025–2026, IEEE ICME 2025–2026, IEEE ICASSP 2026, ACM Multimedia Asia 2026.
- **Tutorial / challenge leadership:** ACM Multimedia 2025 and 2026 tutorials; IEEE ICME 2026 and IEEE IVMSIP 2026 tutorials; VQualA challenge at ICCV 2025 and IEEE SPS Challenge Program 2026; multimedia quality challenges/workshops at ACM Multimedia 2026.
- **IEEE UK & Ireland:** Chair, Election Committee, IEEE UK & Ireland Signal Processing Society Chapter.

Selected Publications

- **Blind Stereoscopic Omnidirectional Image Quality Assessment Using Predictive Coding Hierarchy**
Wei Zhou, André Kaup. IEEE Transactions on Multimedia, 2026.
- **LIQA: Lifelong Blind Image Quality Assessment**
Jianzhao Liu*, Wei Zhou*, Xin Li, Jiahua Xu, Zhibo Chen. IEEE Transactions on Multimedia, 2022. ESI Highly Cited Paper.
- **Blind Quality Assessment of Dense 3D Point Clouds with Structure Guided Resampling**
Wei Zhou, Qi Yang, Wu Chen, Qiuping Jiang, Guangtao Zhai, Weisi Lin. ACM Transactions on Multimedia Computing, Communications, and Applications, 2024.
- **Perceptual Depth Quality Assessment of Stereoscopic Omnidirectional Images**
Wei Zhou, Zhou Wang. IEEE Transactions on Circuits and Systems for Video Technology, 2024.
- **No-Reference Quality Assessment for 360-degree Images by Analysis of Multifrequency Information and Local-global Naturalness**
Wei Zhou, Jiahua Xu, Qiuping Jiang, Zhibo Chen. IEEE Transactions on Circuits and Systems for Video Technology, 2022.
- **Tensor Oriented No-Reference Light Field Image Quality Assessment**
Wei Zhou, Likun Shi, Zhibo Chen, Jinglin Zhang. IEEE Transactions on Image Processing, 2020.
- **Dual-Stream Interactive Networks for No-Reference Stereoscopic Image Quality Assessment**
Wei Zhou, Zhibo Chen, Weiping Li. IEEE Transactions on Image Processing, 2019.
- **VQM4HAS: A Real-time Quality Metric for HEVC Videos in HTTP Adaptive Streaming**
Hadi Amirpour, Jingwen Zhu, Wei Zhou, Patrick Le Callet, Christian Timmerer. IEEE Transactions on Multimedia, 2025.
- **Perception-oriented Bidirectional Attention Network for Image Super-Resolution Quality Assessment**
Yixiao Li, Xiaoyuan Yang, Guanghui Yue, Jun Fu, Qiuping Jiang, Xu Jia, Paul Rosin, Hantao Liu, Wei Zhou. IEEE Transactions on Image Processing, 2025.

Publication record. More than 160 research papers, including over 70 papers in IEEE/ACM Transactions and ~10 IEEE T-MM papers, with sustained T-MM authorship from 2020 to 2026; publications also span IEEE TPAMI, TIP, TCSVT, TNNLS, TMI and ACM TOMM.

Selected Honors and Recognition

- IEEE CASS VSPC Rising Star Honorable Mention Award, 2024.
- IEEE MMSP Rising Star Finalist, 2025.
- ACM SIGMM China Outstanding Doctoral Dissertation Award, 1st place, 2022.
- CVPR CLIC Perceptual Metric Track Grand Challenge Winner, 1st place, 2021; IEEE VCIP Live Broadcasting Video Quality Assessment Challenge Winner, 1st place, 2025.
- Stanford University World Top 2% Scientists, 2024 and 2025; World Top 0.5% Scholar in Digital Image Processing (2025 ranking).
- Exceptional Talent endorsed by the Royal Society, 2023.

Research Expertise Relevant to T-MM

- **Perceptual multimedia quality:** subjective/objective image and video quality assessment, quality of experience, perceptual fidelity and naturalness, saliency and viewing behavior.
- **Immersive and 3D media:** 360-degree/VR, stereoscopic and spatial video, point clouds, light fields, 3D Gaussian splatting, depth perception, and quality modelling for emerging visual representations.
- **Multimodal and generative AI:** multimodal learning, foundation-model-assisted evaluation, AI-generated media quality, spatial intelligence, and embodied AI evaluation.
- **Multimedia systems and optimization:** quality-aware coding, adaptive streaming, rate-quality trade-offs, and perceptual optimization of multimedia processing pipelines.
- **Application domains:** healthcare multimedia, computational imaging, human-centric AI, affective computing, education, and autonomous/robotic visual systems.

Education and International / Industry Experience

- PhD, University of Science and Technology of China, 2021; joint PhD research at the University of Waterloo, Canada (2019–2021) with Prof. Zhou Wang (FIEEE, FRSC, FCAE, Canada Research Chair); outstanding doctoral dissertation recognition.
- Postdoctoral Researcher, University of Waterloo, Canada, 2021–2023.
- **Research / industry experience:** National Institute of Informatics (Japan), Microsoft Research, Alibaba, and Intel.
- **Visiting appointments:** Dalian University of Technology (2024), Shenzhen University (2025), Technical University of Munich (2026).

Selected Multimedia Community Activities

- **General Chair, IEEE QoMEX 2026, Cardiff;** Steering Committee Member, QoMEX.
- Lead Tutorial Organizer & Speaker, ACM Multimedia 2025–2026, IEEE ICME 2026, and IEEE IVMSP 2026.
- Lead Workshop Organizer, Multimedia Computing for Health and Medicine (MCHM), ACM Multimedia 2025–2026.
- Grand Challenge / benchmark organizer: VQualA (ICCV 2025; IEEE SPS Challenge Program 2026), Machine-oriented Visual Media Quality Assessment (ACM Multimedia 2026), Scientific Image Quality Assessment (IEEE ICME 2026).
- Special-session organizer across IEEE ICME, ICASSP, ICIP, VCIP, QoMEX, and ACM Multimedia Asia, covering generative AI evaluation, multimedia quality, embodied multimedia, and multimodal medical AI.
- Peer-review recognition includes IEEE VCIP Best Reviewer Award (2021) and Journal of Electronics & Information Technology Outstanding Reviewer Awards (2024 and 2025).

T-MM / IEEE Editorial Readiness

- Current member of the IEEE CASS Multimedia Systems and Applications Technical Committee and elected member of the IEEE SPS Multimedia Signal Processing Technical Committee.
- IEEE Senior Member with sustained service across the two IEEE co-sponsoring communities most directly aligned with T-MM.
- Approximately 10 IEEE T-MM papers, with sustained T-MM authorship from 2020 to 2026, including first/co-first-author work and a 2026 first-author paper on stereoscopic omnidirectional image quality assessment.
- T-MM reviewing record: reviewed 20+ manuscripts for IEEE Transactions on Multimedia over the past three years, complemented by extensive manuscript-assessment experience as Senior Area Chair / Area Chair at IEEE ICME, IEEE ICIP, IEEE ICASSP and ACM Multimedia.
- General Chair of IEEE QoMEX 2026 and Steering Committee Member of QoMEX, demonstrating sustained leadership in multimedia quality and quality-of-experience research.
- Standards and research-quality experience through service on the British Standards Institution and EPSRC Peer Review College.
- UK/Europe-based academic with prior research and industry experience across Germany, Canada, Japan and China, including Microsoft Research, Alibaba and Intel.
- Research portfolio spans perceptual multimedia quality, immersive/3D media, multimodal and generative AI, multimedia systems, streaming, quality of experience, and human-centric multimedia intelligence.
- Invited engagement includes a talk at Meta (2025) and a VQEG seminar on quality assessment of AI-generated content (2025).
- First-/co-first-author contributions span stereoscopic, omnidirectional/VR, light-field, point-cloud, super-resolution, and enhanced visual content, demonstrating broad technical coverage within T-MM's scope.
- Recent research trajectory extends perceptual multimedia evaluation toward spatial video, 3D Gaussian splatting, multimodal/foundation-model-assisted quality assessment, and generative-media reliability.
- Recent tutorial leadership includes ACM Multimedia 2026 on Perceptual Reliability in Multimedia, IEEE ICME 2026 on Generative AI Benchmark and Evaluation, and IEEE IVMSP 2026 on Perceptual Visual Signal Processing.
- International collaborations in perceptual quality, multimedia systems, multimodal AI, and immersive visual computing span Europe, Asia, and North America.
- Current research and service bridge established multimedia quality assessment with emerging generative, immersive, and embodied media topics relevant to T-MM.
- Experience spans authoring, reviewing, editorial handling, technical-program leadership, tutorials, challenges, and standards activities across the multimedia research lifecycle.
- Technical breadth covers perception, representation, learning, coding/streaming, and quality of experience, supporting manuscript assessment across multiple T-MM areas.